

conditional heteroscedasticity (GARCH) model introduced by Bollerslev (1986), and an HMA-VIX adjustment model that combines the stock's current realized volatility with an implied volatility measure¹.

We fit the parameters of these models to the SP500 and R2000 indexes over the one-year period July 1, 2011 through June 30, 2012 and compare the performance of the models. Readers are encouraged to experiment with these techniques to determine which model work best for their needs.

Price Returns

Price returns for each index were computed as the natural log of close-to-close price change:

$$\gamma_t = \ln(p_t/p_{t-1}) \quad (6.13)$$

A general short-term model of price returns is:

$$\gamma_t = C + \sigma_t \varepsilon_t \quad (6.14)$$

where C is a constant, ε_t is random noise with distribution $\varepsilon_t \sim N(0, 1)$, and σ_t is the time varying volatility component. In practice, the short-term constant term C is rarely known in advance and analysts often use a simplifying assumption of $C = 0$. Then the general short-term price returns model simplifies to:

$$\gamma_t = \sigma_t \varepsilon_t \quad (6.15)$$

Data Sample We used the SP500 and R2000 index returns over the one-year period July 1, 2011 through June 30, 2012 as our data sample. Figure 6.3a shows the daily price returns for the SP500. The standard deviation of daily returns over this period was 1.5% and the SP500 index had price swings as high as +4.6% on August 9, 2011 and as low as -6.9% on August 8, 2011. These large swings occurred during the US debt crisis (Aug.–Sep. 2011). Figure 6.3b shows the price returns for the R2000 index over the same period. The R2000 index had a standard deviation of daily returns of 2.1%, and price swings as high as +6.7% on August 9, 2011 and as low as -9.3% on August 8, 2011. As is evident in both figures, volatility was fairly low through middle of August 2011 when there was a sudden spike in volatility that caused it to remain high through year

¹The HMA-VIX volatility model was presented at Curt Engler's CQA/SQA Trading Seminar (February 2009), "Volatility: Is it safe to get back in the water?" and taught as part of the volatility section in Cornell University's Graduate Financial Engineering Program, "Introduction to Algorithmic Trading," Fall 2009 (Kissell and Malamut). The HMA-VIX model was also published in Journal of Trading, "Intraday Volatility Models: Methods to Improve Real-Time Forecasts," Fall 2012.